Human influences on ecosystems

Mark scheme and worked solutions · 30 questions

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Question 1

core · 1 mark · 0610_T20_P1_20261018_Q1

Answer: B

Rationale
Herbicides kill weeds and reduce competition for light, water and minerals. Fertilisers supply mineral ions; herbicides are not nutrients.

Question 2

core · 1 mark · 0610_T20_P1_20261018_Q2

Answer: D

Rationale
A monoculture concentrates one crop and can allow pests or diseases to spread efficiently. It can make mechanised management easier, not impossible.

Question 3

core · 1 mark · 0610_T20_P1_20261018_Q3

Answer: A

Rationale
High stocking density can facilitate disease spread. Intensive systems still require feed, produce waste and raise welfare considerations.

Question 4

core · 1 mark · 0610_T20_P1_20261018_Q4

Answer: C

Rationale
Biodiversity here means the number of different species. Abundance of one species and environmental measurements are different quantities.

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Question 5

core · 1 mark · 0610_T20_P1_20261018_Q5

Answer: D

Rationale
Clearing woodland removes habitat. Counting, seed banking and protection can support conservation rather than directly destroy it.

Question 6

core · 1 mark · 0610_T20_P1_20261018_Q6

Answer: B

Rationale
Removing trees reduces binding by roots and exposes soil to erosion. The alternatives do not describe a mechanism of soil loss.

Question 7

core · 1 mark · 0610_T20_P1_20261018_Q7

Answer: C

Rationale
Decomposers respire aerobically while breaking down organic material, using dissolved oxygen. Fish do not photosynthesise and oxygen is not converted into sunlight.

Question 8

core · 1 mark · 0610_T20_P1_20261018_Q8

Answer: A

Rationale
Non-biodegradable plastic persists and may cause ingestion or entanglement. It is neither necessarily soluble nor harmless food.

Question 9

core · 1 mark · 0610_T20_P1_20261018_Q9

Answer: B

Rationale
Methane and carbon dioxide absorb outgoing infrared radiation and contribute to warming. The other listed pairs are not the principal greenhouse gases described here.

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Question 10

core · 1 mark · 0610_T20_P1_20261018_Q10

Answer: C

Rationale
A sustainable resource is produced as rapidly as it is removed, so it does not run out. The other practices deplete the resource.

Question 11

core · 1 mark · 0610_T20_P1_20261018_Q11

Answer: D

Rationale
Seed banks preserve viable plant material for future propagation. They complement habitat protection rather than require habitat removal or destruction of seeds.

Question 12

supplement · 1 mark · 0610_T20_P1_20261018_Q12

Answer: A

Rationale
Larger mesh allows small fish to escape, grow and potentially breed, helping replenish stocks. It does not eliminate all predators or stop reproduction.

Question 13

supplement · 1 mark · 0610_T20_P1_20261018_Q13

Answer: C

Rationale
Replanting helps replace harvested trees. The other measures reduce forest regeneration or convert land to another use.

Question 14

supplement · 1 mark · 0610_T20_P1_20261018_Q14

Answer: A

Rationale
Artificial insemination introduces sperm into the female reproductive tract and can facilitate planned crosses. It is sexual reproduction, not cloning.

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Question 15

supplement · 1 mark · 0610_T20_P1_20261018_Q15

Answer: B

Rationale
Low variation can leave few resistant or tolerant variants when conditions change. It does not guarantee universal adaptation or eliminate disease.

Question 16

core · 4 marks · 0610_T20_P1_20261018_Q16

(a) [2]

- Machinery allows larger areas to be cultivated or harvested efficiently. (1)
- Fertiliser supplies mineral ions needed for plant growth. (1)

(b) [2]

- Insecticide reduces damage by insect pests. (1)
- Herbicide reduces competition with weeds for resources. (1)

Worked solution
(a) Machinery allows larger areas to be cultivated or harvested efficiently. Fertiliser supplies mineral ions needed for plant growth.

(b) Insecticide reduces damage by insect pests. Herbicide reduces competition with weeds for resources.

Question 17

core · 4 marks · 0610_T20_P1_20261018_Q17

(a) [2]

- A single crop can be managed and harvested efficiently with machinery. (1)
- A pest or disease of that crop can spread through many plants. (1)

(b) [2]

- Controlled conditions and reduced movement can increase production efficiency. (1)
- Crowding can increase disease spread or cause welfare problems. (1)

Worked solution
(a) A single crop can be managed and harvested efficiently with machinery. A pest or disease of that crop can spread through many plants.

(b) Controlled conditions and reduced movement can increase production efficiency. Crowding can increase disease spread or cause welfare problems.

Question 18

core · 4 marks · 0610_T20_P1_20261018_Q18

(a) [2]

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- There are fewer suitable breeding sites. (1)
- Reduced reproduction can cause the population to decline. (1)

(b) [2]

- Predators of the fish may lose an important food source. (1)
- Organisms eaten by the fish may increase when predation is reduced. (1)

Worked solution
(a) There are fewer suitable breeding sites. Reduced reproduction can cause the population to decline.

(b) Predators of the fish may lose an important food source. Organisms eaten by the fish may increase when predation is reduced.

Question 19

core · 4 marks · 0610_T20_P1_20261018_Q19

(a) [2]

- Roots no longer bind the soil as effectively. (1)
- Reduced interception and uptake allow more rapid surface runoff towards rivers. (1)

(b) [2]

- Fewer trees remove carbon dioxide by photosynthesis. (1)
- Burning or decomposition of cleared wood releases carbon dioxide. (1)

Worked solution
(a) Roots no longer bind the soil as effectively. Reduced interception and uptake allow more rapid surface runoff towards rivers.

(b) Fewer trees remove carbon dioxide by photosynthesis. Burning or decomposition of cleared wood releases carbon dioxide.

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Question 20

core · 4 marks · 0610_T20_P1_20261018_Q20

(a) [2]

- The decrease is 48 − 18 = 30 species. (1)
- Biodiversity has decreased because fewer different species remain. (1)

(b) [2]

- A species may exist only in that habitat. (1)
- If all its individuals lose the conditions needed to survive and reproduce, none may remain anywhere. (1)

Worked solution
(a) The decrease is 48 − 18 = 30 species. Biodiversity has decreased because fewer different species remain.

(b) A species may exist only in that habitat. If all its individuals lose the conditions needed to survive and reproduce, none may remain anywhere.

Question 21

supplement · 4 marks · 0610_T20_P1_20261018_Q21

(a) [2]

- Extra nitrate and other ions increase producer growth. (1)
- When more algae die, decomposers have more dead material to break down. (1)

(b) [2]

- Aerobic respiration by decomposers uses dissolved oxygen. (1)
- Low dissolved oxygen prevents fish obtaining enough oxygen for respiration. (1)

Worked solution

(a) Extra nitrate and other ions increase producer growth. When more algae die, decomposers have more dead material to break down.

(b) Aerobic respiration by decomposers uses dissolved oxygen. Low dissolved oxygen prevents fish obtaining enough oxygen for respiration.

Question 22

core · 4 marks · 0610_T20_P1_20261018_Q22

(a) [2]

- Animals may become entangled, restricting movement or causing injury. (1)
- Ingested plastic may block the gut or reduce feeding. (1)

(b) [2]

- Decomposers do not readily break it down. (1)
- Water or wind can transport persistent plastic to other habitats. (1)

Worked solution

(a) Animals may become entangled, restricting movement or causing injury. Ingested plastic may block the gut or reduce feeding.

(b) Decomposers do not readily break it down. Water or wind can transport persistent plastic to other habitats.

Question 23

core · 4 marks · 0610_T20_P1_20261018_Q23

(a) [2]

- Fossil-fuel combustion releases carbon dioxide. (1)
- Cattle farming releases methane. (1)

(b) [2]

- They absorb more outgoing infrared radiation and reduce heat loss to space. (1)
- The enhanced greenhouse effect can raise global temperatures and alter climate. (1)

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Worked solution
(a) Fossil-fuel combustion releases carbon dioxide. Cattle farming releases methane.

(b) They absorb more outgoing infrared radiation and reduce heat loss to space. The enhanced greenhouse effect can raise global temperatures and alter climate.

Question 24

core · 4 marks · 0610_T20_P1_20261018_Q24

(a) [2]

- They are replenished by growth or reproduction. (1)
- Their removal must not exceed their replenishment over time. (1)

(b) [2]

- Replant or regenerate harvested forest. (1)
- Limit fish catches or protect breeding periods. (1)

Worked solution
(a) They are replenished by growth or reproduction. Their removal must not exceed their replenishment over time.

(b) Replant or regenerate harvested forest. Limit fish catches or protect breeding periods.

Question 25

core · 4 marks · 0610_T20_P1_20261018_Q25

(a) [2]

- The predator increases mortality of birds or their young. (1)
- If deaths exceed successful reproduction, the population decreases. (1)

(b) [2]

- Control the introduced predator and protect the bird's habitat. (1)
- Use monitored captive breeding or protected breeding sites to increase survival and reproduction. (1)

Worked solution
(a) The predator increases mortality of birds or their young. If deaths exceed successful reproduction, the population decreases.

(b) Control the introduced predator and protect the bird's habitat. Use monitored captive breeding or protected breeding sites to increase survival and reproduction.

Question 26

supplement · 4 marks · 0610_T20_P1_20261018_Q26

(a) [2]

- Quotas limit the rate of tree removal. (1)
- Replanting replaces harvested trees as the forest grows. (1)

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(b) [2]

- Protected areas maintain habitats where logging is restricted. (1)
- Education encourages people to understand and support sustainable use. (1)

Worked solution
(a) Quotas limit the rate of tree removal. Replanting replaces harvested trees as the forest grows.

(b) Protected areas maintain habitats where logging is restricted. Education encourages people to understand and support sustainable use.

Question 27

supplement · 4 marks · 0610_T20_P1_20261018_Q27

(a) [2]

- Closed seasons allow fish to reproduce with less fishing pressure. (1)
- Quotas limit the number or mass removed from the stock. (1)

(b) [2]

- Monitoring shows whether stocks recover and whether limits need adjustment. (1)
- Suitable larger mesh lets more immature fish escape and later breed. (1)

Worked solution
(a) Closed seasons allow fish to reproduce with less fishing pressure. Quotas limit the number or mass removed from the stock.

(b) Monitoring shows whether stocks recover and whether limits need adjustment. Suitable larger mesh lets more immature fish escape and later breed.

Question 28

supplement · 4 marks · 0610_T20_P1_20261018_Q28

(a) [2]

- It reduces the risk of species becoming extinct. (1)
- It retains genetic resources that may provide useful food, drugs or other products. (1)

(b) [2]

- It helps maintain nutrient cycling by interacting organisms. (1)
- It protects resource provision and the relationships supporting the community. (1)

Worked solution
(a) It reduces the risk of species becoming extinct. It retains genetic resources that may provide useful food, drugs or other products.

(b) It helps maintain nutrient cycling by interacting organisms. It protects resource provision and the relationships supporting the community.

Question 29

supplement · 4 marks · 0610_T20_P1_20261018_Q29

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(a) [2]

- Sperm from a selected male can be collected and transferred to another site. (1)
- The sperm can be introduced into a selected female's reproductive tract. (1)

(b) [2]

- Eggs are fertilised with sperm outside the female's body. (1)
- The resulting embryos are transferred to a female for development. (1)

Worked solution
(a) Sperm from a selected male can be collected and transferred to another site. The sperm can be introduced into a selected female's reproductive tract.

(b) Eggs are fertilised with sperm outside the female's body. The resulting embryos are transferred to a female for development.

Question 30

supplement · 4 marks · 0610_T20_P1_20261018_Q30

(a) [2]

- There may be few alleles conferring resistance to a new disease. (1)
- There may be few variants suited to a changed climate or habitat. (1)

(b) [2]

- It preserves a wider range of inherited features. (1)
- This improves the chance that some offspring can survive future environmental challenges. (1)

Worked solution
(a) There may be few alleles conferring resistance to a new disease. There may be few variants suited to a changed climate or habitat.

(b) It preserves a wider range of inherited features. This improves the chance that some offspring can survive future environmental challenges.

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